

A1: Floe-Property and Domain Sweep

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For which floes does the transmissibility inverse work?

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SHARC buoy project Code: `runA1FloePropertySweep.m`, commit 0389536 Results: `Results/A1/A1_20261008_180633`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 3 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). Settings unchanged; all 24 scenarios recomputed (checkpoints of the 3 Oct run set aside). This summary supersedes A1_20261003_091530. The higher signal level lowers σ about $7\times$ and moves both domain boundaries.

Summary

Question. With the sensing held fixed (Level 2C at fractions of each floe’s true radius, reference s26, LSM6DSV16X, setting F), which floes can the inverse identify? 24 pre-registered scenarios: 8 one-parameter ring cases, 4 corners, 12 physical floes (basin, tank, SHARC model floe, MIZ). **Outcome.** 16 Identifiable, 1 Marginal, 1 Model check failed and 6 Not identifiable (class robust to the EMM truncation). The domain still has **two boundaries with different causes**. *Lower, physical:* the stiffness is lost as R/ℓ_f falls below about 1 ($\sigma_{\ln \beta}$ 1.75% at $R/\ell_f = 0.96$, 13 to 15% at 0.57 to 0.65, 44 to 76% for the model floe) while the radius stays measurable ($\sigma_{\ln R} \leq 0.24\%$). *Upper, estimator-limited:* at $kR \approx 3.7$ the precision is the best in the study ($\sigma_{\ln \beta} \approx 0.04\%$) but the Welch bias limits the class (d_W 0.67 and 1.07 for two of the three large floes). Between them, kR sets the precision. Stiffness itself matters little ($\sigma_{\ln \beta}$ 0.050 to 0.074% for β/β_0 from 0.25 to 4). The model floe is effectively rigid ($\sigma_{\ln \beta}/\sigma_{\ln R} = 600$ to 860). The EMM truncation bias of the [50 10 10] model is now large in every scenario, which makes V1 decisive for real data.

1 What the study does

Scaling. Each physical floe is converted with one water depth throughout: $D = Eh^3/(12(1 - \nu^2))$, $\beta = D/(\rho_w g H^4)$, $\gamma = \rho_i h/(\rho_w H)$, $R = R_{\text{phys}}/H$, with γ known and only $\theta = [\ln \beta, \ln R]$ estimated. Basin and tank floes use $H = 1.88$ m and the validated basin band (3.178 to 8.301 rad/s, $\alpha = 1.95$ to 13.15). The four MIZ floes use the effective depth of `screenFloeRegime` (122 m) and are evaluated *Froude-scaled* to the basin: the EMM sees only α , β , γ , R , so the 6 to 12.5 s band maps to basin frequencies, cut to the basin band. The equivalent full-scale experiment is $H_s = 3.25$ m, $T_p = 10.1$ s, 880 min, $\Delta t = 0.81$ s. Poisson’s ratio is passed to the EMM (0.40 for polypropylene).

Sensing held fixed. Sensors at the Level 2C fractions of each true R , converted once and held fixed while $\ln R$ is differentiated. The layout is *not* re-optimised per floe, so a change in performance comes from the floe, not from the layout.

Model validity before identifiability. Each scenario is checked before its class is read: (i) node grid 0.2 against 0.1 rad/s, feature error projected with the scenario’s own F , $d < 0.1$, otherwise the scenario is evaluated on the 0.1 rad/s grid (pre-registered fallback); (ii) truncation [50 10 10] against [70 15 15] as *metric* convergence: σ within 1%, d_W within 0.02, same class; (iii) all bins inside the validated α window. A failure is labelled *Model check failed*; where the class is identical at both truncations the reporting layer adds “class robust” (N.I.[†]). The feature-level truncation bias d_{trunc} is reported for V1 but is not a gate.

2 Results

Parameter space (basin, $\gamma = \gamma_0$, $\nu = 0.3$; I = Identifiable, M = Marginal; * evaluated on the 0.1 rad/s node grid after the fallback).

Scenario	β/β_0	R/R_0	kR_{pk}	R/ℓ_f	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	d_W	$\kappa(F)$	d_{trunc}	Class
ring β	0.25	1	1.84	6.54	0.074%	0.012%	0.280	129	4.6	I
ring β	0.5	1	1.84	5.50	0.054%	0.010%	0.382	87	5.5	I
p_0	1	1	1.84	4.63	0.050%	0.008%	0.436	83	5.2	I
ring β	2	1	1.84	3.89	0.053%	0.009%	0.445	114	4.5	I
ring β	4	1	1.84	3.27	0.062%	0.012%	0.420	142	3.1	I
ring R	1	0.5	0.92	2.31	0.131%	0.034%	0.114	90	3.1	I
ring R	1	0.75	1.38	3.47	0.077%	0.015%	0.337	168	5.3	I
ring R	1	1.5	2.75	6.94	0.044%	0.007%	0.332	86	3.0	I
ring R^*	1	2	3.67	9.25	0.042%	0.006%	0.667	121	21	M (bias)
corner	0.5	0.5	0.92	2.75	0.147%	0.037%	0.170	199	4.8	I
corner*	0.5	2	3.67	11.0	0.049%	0.005%	1.066	161	25	N.I. [†] (bias)
corner	2	0.5	0.92	1.95	0.126%	0.030%	0.098	41	1.4	I
corner*	2	2	3.67	7.78	0.038%	0.006%	0.291	87	7.7	I

Large- R corner with $\beta_0/2$: Not identifiable through $d_W > 1$ at both truncations; its σ moves by 1.0% between truncations, exactly at the gate, so the run labels it Model check failed.

Physical floes (N.I.[†]: Not identifiable at both truncations; $\sigma_{\ln \beta}$ in brackets is not converged and not quotable).

Scenario	Description	kR_{pk}	R/ℓ_f	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	d_W	d_{trunc}	Class
pp3mm_R072	PP 3 mm, $R = 0.72$ m (p_0 analogue)	1.84	5.06	0.049%	0.009%	0.420	5.6	I
pp5mm_D100	PP 5 mm, 1 m diameter	1.27	2.40	0.088%	0.022%	0.167	3.7	I
pp5mm_D40	PP 5 mm, 40 cm	0.51	0.96	1.754%	0.080%	0.115	2.8	M; check failed [‡]
pp10mm_D40	PP 10 mm, 40 cm	0.51	0.57	(14.5%)	0.090%	0.098	3.1	N.I. [†]
modelFloe	SHARC model floe, $E = 0.5/1.5/3$ GPa [§]	0.56	0.19–0.29	(44–76%)	0.07–0.11%	0.13	15–18	N.I. [†]
miz_R10_h1	$R = 10$ m, $h = 1$ m, 6 GPa	0.39	0.65	(12.8%)	0.236%	0.031	5.1	N.I. [†]
miz_R25_h1	$R = 25$ m, $h = 1$ m, 6 GPa	0.98	1.64	0.141%	0.028%	0.092	10.5	I
miz_R50_h1	$R = 50$ m, $h = 1$ m, 6 GPa	1.96	3.27	0.058%	0.011%	0.428	4.7	I
miz_R25_h05_E3	$R = 25$ m, $h = 0.5$ m, 3 GPa	0.98	3.27	0.135%	0.030%	0.219	7.7	I

[‡]Marginal ($\sigma > 1\%$) at both truncations; σ moves by 1.3%, so the metric gate fails. [§]Provisional: 4 cm volume-equivalent plate of the 44 cm, 6 cm lens, E unknown. The basin p_0 row is in the parameter table and reproduces the corrected E1 headline exactly.

Model checks. Node grid: 21 pass at 0.2 rad/s; the three $R = 2R_0$ cases ($kR = 3.67$) exceed $d = 0.1$ ($d = 0.139, 0.126, 0.102$) and are evaluated on the 0.1 rad/s grid, as pre-registered. Before the amendment all 24 passed (largest $d = 0.023$): d is projected with F , which grew with the signal level. α : all inside [1.70, 13.85]. Truncation metric gate: 17 pass (largest change in σ 0.9%, in d_W 0.0011); 7 fail, all small or stiff floes plus the large- R , $\beta_0/2$ corner. Feature-level truncation bias: $d_{trunc} \geq 1.35$ in **all 24** scenarios (largest 25 at the large- R corner; model floes 15 to 18).

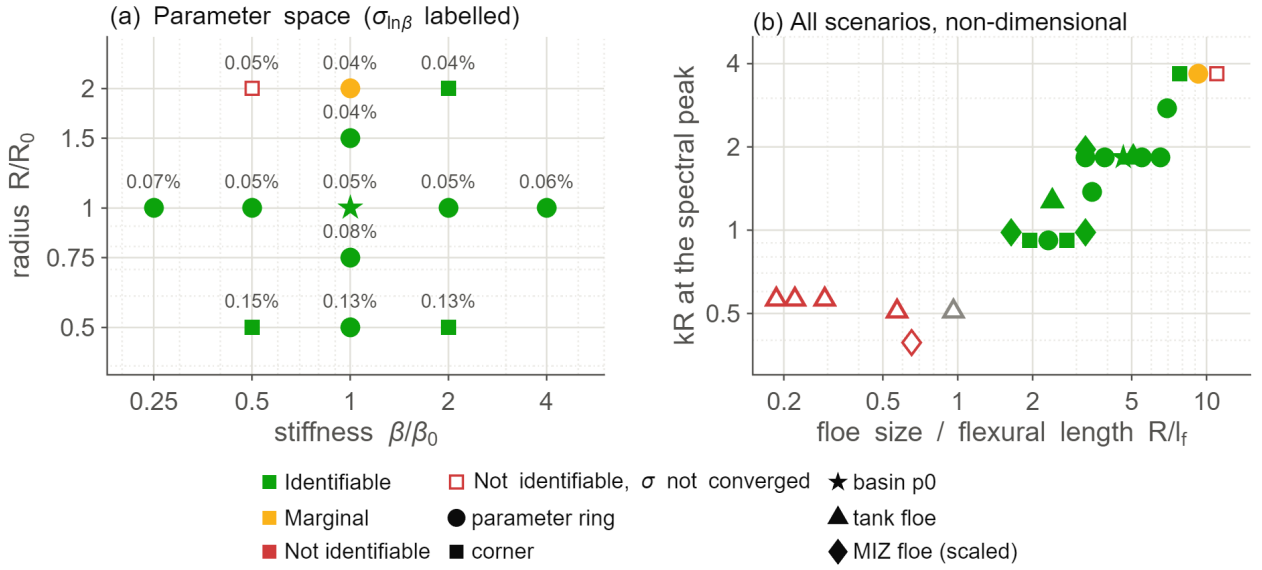


Figure 1. (a) Parameter-space scenarios, labelled with $\sigma_{\ln \beta}$. (b) All scenarios in the non-dimensional plane: floe size over flexural length against kR at the spectral peak. Colour: class; marker: group; open red: Not identifiable with σ not converged.

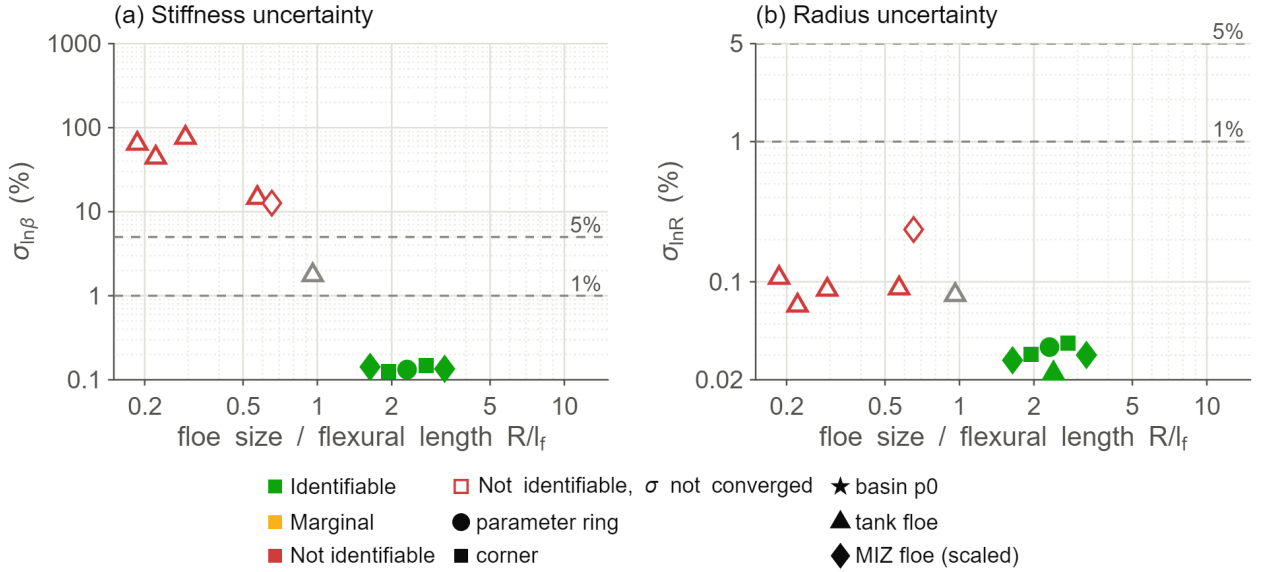


Figure 2. Stiffness (a) and radius (b) uncertainty against R/ℓ_f . Dashed: the 1% and 5% class limits. Below $R/\ell_f \approx 1$ the stiffness is lost; the radius is not.

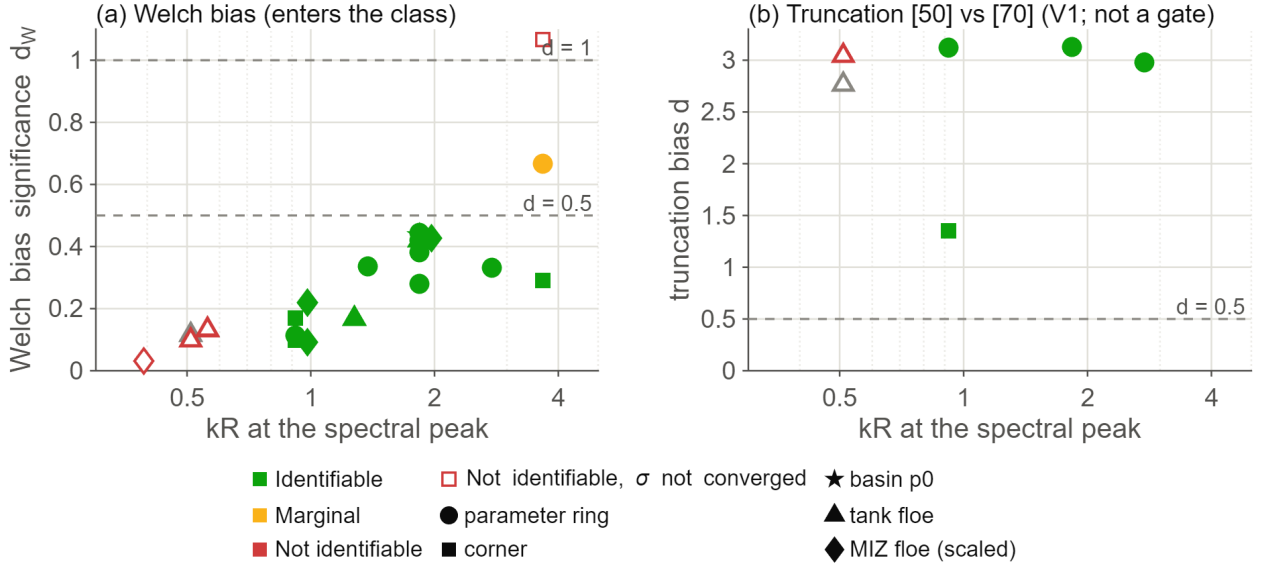


Figure 3. (a) Welch bias significance, which enters the class, against kR : it grows with kR and exceeds 0.5 for two of the three largest floes. (b) Feature-level bias of the [50 10 10] model against [70 15 15] (information for V1, not a gate).

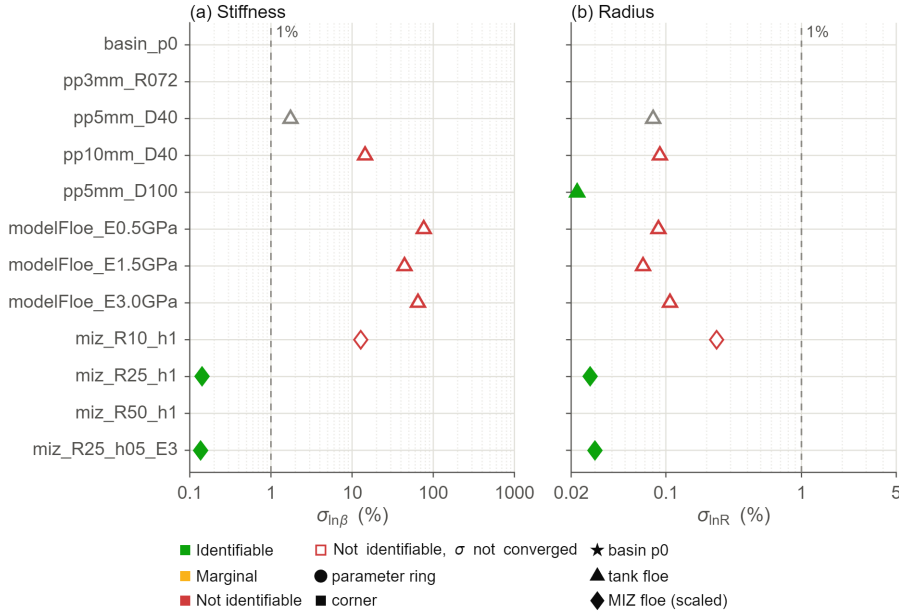


Figure 4. The physical floes: stiffness (a) and radius (b) uncertainty. Open markers: Not identifiable, value not converged.

3 Interpretation

- Lower boundary: the floe must be comparable to or larger than its flexural length.** $\sigma_{\ln \beta}$ is at most 0.15% for every scenario with $R/\ell_f \geq 1.64$, 1.75% at $R/\ell_f = 0.96$, 13 to 15% at 0.57 to 0.65, and 44 to 76% for the model floe (0.19 to 0.29). The 5% (Marginal) limit is crossed between $R/\ell_f = 0.65$ and 0.96, the 1% (Identifiable) limit between 0.96 and 1.64; neither is located, only bracketed. A floe much smaller than ℓ_f hardly bends in the band, so its motion carries little stiffness information; its size still shapes the transmissibility, so R stays measurable everywhere. The amendment moved the boundary down: at $R/\ell_f = 0.96$ $\sigma_{\ln \beta}$ was 14% before and is 1.75% now.
- Upper boundary: estimator-limited, and it depends on the stiffness.** At $kR = 3.67$ the Fisher precision is the best in the study (0.038 to 0.049%), but the Welch bias limits the class: $d_W = 0.667$ (Marginal) at β_0 , 1.066 (Not identifiable) at $\beta_0/2$, and only 0.291 (Identifiable) at $2\beta_0$. A softer floe has a shorter flexural wavelength, so its transmissibilities vary faster in frequency and the window smooths them more. These cases also need the finer 0.1 rad/s model grid. This is the mechanism of D2, where $L = 4096$ removes most of the Welch bias; whether it recovers these floes is a hypothesis for a logged diagnostic, not a result of A1.
- In between, kR sets the precision.** $\sigma_{\ln \beta}$ is 0.13 to 0.15% at $kR \approx 0.9$ to 1, 0.08 to 0.09% at 1.3, 0.05 to 0.07% at 1.8 and 0.04% at 2.8 to 3.7. R/ℓ_f alone does not collapse the results: miz_R25_h05.E3 and miz_R50_h1 share $R/\ell_f = 3.27$ but have $kR = 0.98$ and 1.96 and $\sigma_{\ln \beta} = 0.135\%$ and 0.058%. R/ℓ_f decides whether flexural information exists; kR decides how much the sensors see.
- Stiffness itself is benign.** Over a 16-fold range of β at R_0 , $\sigma_{\ln \beta}$ stays between 0.050 and 0.074%, smallest near β_0 to $2\beta_0$. d_W is not monotone (0.28 to 0.45, largest near β_0 to $2\beta_0$), so the bias budget at p_0 is near its tightest.
- Tank floes.** The 3 mm, 0.72 m polypropylene disc reproduces p_0 (0.049% against 0.050%), a check of the non-dimensional framework, and the 1 m disc is comfortably Identifiable (0.088%). The 5 mm, 40 cm disc is now

- Marginal (1.75%) instead of Not identifiable, but its σ is not converged in truncation. The SHARC model floe is not identifiable in stiffness under any assumed E : **the recorded hypothesis is confirmed**, the floe is effectively rigid in the basin band ($\sigma_{\ln \beta}/\sigma_{\ln R} = 600$ to 860). A basin stiffness experiment needs a thin plate of about 1 m diameter.
6. **MIZ (scaled screening)**. At $h = 1$ m, $E = 6$ GPa: 10 m Not identifiable, 25 m Identifiable (0.141%; it was on the boundary at 1.005% before the amendment), 50 m Identifiable. A thinner, softer 25 m floe is Identifiable at the same kR (0.135%).
 7. **The six robust Not identifiable cases**. Five have weak stiffness information and an ill-conditioned weak direction (κ from 2.9×10^3 to 1.4×10^6); their class does not depend on the truncation, their $\sigma_{\ln \beta}$ does, so only the class is reported. The sixth, the large- R corner at $\beta_0/2$, is precise but biased ($d_W > 1$).
 8. **Converged metrics do not mean a converged model; this is now the dominant caveat**. With the corrected signal level the [50 10 10] model differs from [70 15 15] by $d_{\text{trunc}} = 1.35$ to 25 in *every* scenario (16 of 24 exceeded 0.5 before), although in every scenario that passes the metric gate σ and d_W move by less than 1%. The twin studies are unaffected (data and model share the EMM), but inverting *real* data with [50 10 10] would carry a bias of several standard deviations, most for large kR and for the stiff model floe. V1 (truth truncation [130 20 20]) quantifies this and sets the truncation for real data.

Limitations.

- **Fixed sensing**. Level 2C at fractions of R ; for the 40 cm floes two sensors are 4 cm apart, which 10 cm buoys cannot realise. The small-floe results describe the floe, not a buildable array.
- **One sea state and one direction** (JONSWAP $H_s = 0.05$ m, $\omega_p = 5$ rad/s; R2, V2 and D3 extend this).
- **MIZ cases are Froude-scaled screening**, not a field prediction. A real LSM6DSV16X at full scale is about 8 times quieter in PSD than the scaled case assumes, so the MIZ results are conservative in noise.
- **Fixed basin band**. Changing R changes kR as well; A1 answers an engineering question (which floes work under this excitation band) rather than isolating floe properties.
- **Post-run reporting layer**. The “class robust” note was defined after the first run; the run’s own label (Model check failed) is kept alongside.
- **M1 boundary case**. The pre-registered rule picked miz_R25_h1 because it sat on the Identifiable/Marginal line (1.005%). After the amendment it is comfortably Identifiable, so the rule has to be re-applied to these results before M1 is rerun (logged in `cfg.meta.changes`).

Requirement for the experiment and the thesis

The inverse identifies stiffness for floes with R/ℓ_f of order 1 or more and kR of order 1 or more in the excitation band ($\sigma_{\ln \beta} \leq 0.15\%$ from $R/\ell_f = 1.64$; 1.75% at 0.96); well below $R/\ell_f \approx 1$ only the size can be recovered. For a basin experiment, use a thin plate of about 1 m diameter (3 to 5 mm polypropylene); the SHARC model floe cannot yield a stiffness. For floes with large kR , especially soft ones, the Welch resolution, not the information, limits the class. Before any real-data inversion the EMM truncation must be raised and converged (V1): at [50 10 10] the model error is several standard deviations in every scenario.

Outputs (Results/A1/): A1_20261008_180633.log/.mat/_summary.txt; A1_scenarios.csv, A1_validity.csv, A1_equivalent.csv, A1_reporting.csv; checkpoints/; figures/A1_fig1 to A1_fig4 (.pdf/.png). Superseded: A1_20261003_091530 (pre-amendment; its checkpoints kept as checkpoints_preAlphaFix).